

AdiVaani Phase-II

Building a National AI Ecosystem for Tribal Languages, Knowledge Access, and Cultural Preservation

1. Executive Summary

AdiVaani was initiated to address one of India’s most important but under-served challenges: the absence of robust AI infrastructure for tribal and other low-resource languages. In its first phase, the project focused on building foundational translation capabilities between **Hindi/English and four tribal languages**. This demonstrated that, with the right combination of AI, linguistic collaboration, and community participation, meaningful language technologies can be developed even for extremely low-resource settings.

However, the real need is much broader. Tribal communities do not interact only through Hindi and English. In practice, they engage with **state languages**, local administration, school systems, healthcare services, and public communication systems that often operate in the dominant language of the state. At the same time, a vast amount of knowledge relevant to these communities remains inaccessible because it is either not translated, not digitized, or not available in usable multilingual form.

AdiVaani Phase-II therefore proposes a structured **three-year national program** to move beyond translation and build a **complete AI ecosystem** for tribal languages—covering multilingual translation, speech technologies, OCR, document digitization, knowledge access, cultural preservation, and generative AI. The long-term goal is to create an integrated multilingual platform connecting **English, Hindi, scheduled state languages, tribal languages, and dialectal variants**, while also enabling future **Tribal LLMs** and AI-driven educational, governance, healthcare, and media applications.

2. Vision

The vision of AdiVaani Phase-II is to build **India’s first large-scale AI ecosystem for tribal languages**, serving simultaneously as:

- a language technology mission,
- a knowledge access mission,
- an educational inclusion mission,
- and a cultural preservation mission.

The program aims to create a unified multilingual infrastructure connecting:

English ↔ Hindi ↔ State Languages ↔ Tribal Languages ↔ Dialects

This infrastructure will support multilingual communication, speech and document access, text-book translation, policy access, cultural archiving, and eventually generative AI content creation in tribal languages.

3. Strategic Rationale

In most tribal regions:

- school education is delivered in **state languages**,
- district administration functions in **state languages**,
- health workers communicate in **state languages**,
- and central documentation remains in **Hindi or English**.

The absence of structured **state-language** ↔ **tribal-language bridges** creates major gaps in education, governance, and service delivery. In addition, local knowledge resources such as school textbooks, policy documents, folktales, oral narratives, and cultural material often remain non-digitized and inaccessible to modern AI systems.

Expanding AdiVaani directly addresses these challenges and aligns with broader national priorities around:

- multilingual education,
- digital inclusion,
- last-mile governance,
- cultural preservation,
- and inclusive AI for underserved communities.

4. Core Language and Knowledge Architecture

AdiVaani Phase-II will operate as a layered multilingual and multimodal ecosystem:

English ↔ Hindi ↔ State Languages ↔ Tribal Languages ↔ Dialects

This architecture will support not only translation, but also:

- text-to-text translation,
- speech-to-text and text-to-speech systems,
- near-live speech-to-speech translation,
- OCR for local language printed and scanned material,
- digitization of textbooks and cultural documents,
- multilingual search, summarization, and explanation,
- and future deployment of Tribal LLMs and generative AI tools.

5. Strategic Pillars of the Program

Pillar 1: Foundational AI and Language Infrastructure

This pillar will build the core language technology backbone of the program.

Key activities will include:

- expansion from the current four languages to **15 or more additional tribal languages**,
- integration of **scheduled state languages**,
- support for **dialect-level variation** within tribal languages,
- development of multilingual **translation systems**,
- development of **Automatic Speech Recognition (ASR)** and **Text-to-Speech (TTS)** technologies,
- development of **speech-to-speech translation** systems,
- and large-scale creation of text and speech corpora.

This pillar will also include large-scale collection of:

- parallel translation corpora,
- monolingual corpora,
- speech datasets,
- conversational datasets,
- and dialectal recordings.

Pillar 2: OCR, Digitization, and Knowledge Access Infrastructure

A dedicated pillar will focus on **OCR development and document digitization** for local and tribal language materials.

A large amount of important educational, administrative, and cultural knowledge currently exists only in printed, scanned, or image-based form. Without OCR and digitization, this material cannot be effectively used for search, translation, summarization, learning systems, or future language models.

This pillar will therefore include:

- development of **OCR systems** for tribal and regional scripts where feasible,
- OCR-enabled conversion of **NCERT books and school textbooks** into machine-readable multilingual formats,
- digitization of **state educational materials and local-language policy documents**,
- translation and digitization of **school textbooks into tribal languages**,
- creation of digital repositories of **folktales, oral narratives, songs, stories, and folk cultural material**,
- and preservation of both **written and oral cultural heritage** through structured digital pipelines.

This pillar will serve as a bridge between **printed knowledge** and **AI-enabled access**, while also creating the digital corpus needed for future downstream educational tools and generative AI systems.

Pillar 3: Sectoral Products and Public-Facing Applications

This pillar will focus on real-world applications in priority public sectors.

Education

- multilingual and audio-first learning tools,
- translation and adaptation of school content,
- teacher-assist platforms,
- language learning applications,
- and AI-generated educational content in tribal languages.

Governance

- translation of policy documents and citizen-facing information,
- multilingual welfare scheme explainers,
- support for district-level communication,
- and voice-based access systems for grievance and information support.

Healthcare

- multilingual health communication tools,
- maternal and child health advisories,
- local-language health awareness material,
- and telemedicine language support.

Media and Public Communication

- subtitling of videos,
- dubbing and speech rendering,
- translation of public information material,
- and multilingual dissemination of important announcements.

Pillar 4: Cultural Preservation and Knowledge Heritage

AdiVaani Phase-II will explicitly include a major effort on **cultural preservation**.

This pillar will focus on documenting, digitizing, translating, and preserving:

- folktales,
- oral histories,
- songs and narratives,
- local stories,
- community knowledge systems,
- and culturally important written material.

This effort will ensure that the project is not only solving a present-day access problem, but also creating a long-term **digital knowledge and civilizational archive** for future generations.

Pillar 5: Generative AI and Tribal LLM Development

In the later phase of the project, the data and infrastructure created in the first two years will support the development of a **Tribal Language LLM / Foundation Model**.

This model will go beyond translation and enable a range of **Generative AI functionalities**, including:

- conversational AI,
- content generation in tribal languages,
- summarization and explanation,
- educational content generation,
- policy simplification,
- generation of audio content,
- generation of image-supported educational material,
- and generation of videos and multimedia explainers.

This will position AdiVaani not only as a translation and digitization platform, but also as a **creative and knowledge-generation platform** for tribal languages.

The project will also explore collaboration models with ecosystem partners such as **BharatGen**, **Sarvam AI**, and other relevant organizations for model training, infrastructure support, and deployment.

6. Three-Year Program Plan

Year 1: Language Expansion and Foundational Infrastructure

The first year will focus on:

- expansion to additional tribal languages,
- integration of selected state languages,
- initiation of ASR and TTS development,
- launch of near-live speech-to-speech prototypes,
- large-scale data collection,
- initiation of OCR research and digitization pilots,
- and beginning textbook and policy document digitization efforts.

Year 2: Product Development, OCR Scaling, and Public Deployment

The second year will focus on:

- deployment of translation and speech tools,

- scaling OCR pipelines,
- launching textbook translation and digitization projects,
- building language learning and subtitling systems,
- digitizing cultural and folk content,
- deploying governance and healthcare communication tools,
- and expanding data collection for foundation-model readiness.

Year 3: Tribal LLM and Generative AI Ecosystem

The third year will focus on:

- training a multilingual Tribal LLM / Foundation Model,
- enabling generative AI content creation,
- creating tools for educational content generation,
- generating audio, image-supported, and video-based multilingual content,
- and supporting large-scale applications in education, governance, healthcare, and cultural preservation.

7. Human Resource Requirement

This program is ambitious and cannot be executed meaningfully with a small or conventional project structure. It requires **top-level researchers, top-level developers, and strong interdisciplinary teams**.

The program should therefore be structured around:

- a **core R&D team** led by IIT Delhi for foundational AI, speech, OCR, generative AI, and model development,
- a **dedicated product engineering team** hired at competitive, industry-aligned salaries for deployment and application development,
- a **language and community team** for data collection, validation, annotation, and cultural engagement,
- and a **digitization and documentation team** for OCR, book conversion, and cultural preservation pipelines.

If the project is to be executed in a targeted and nationally meaningful manner, attracting and retaining top talent will be essential.

8. Institutional and Collaboration Framework

AdiVaani Phase-II is envisioned as a nationally relevant platform and will benefit from structured collaboration with:

- state governments for deployment and school-system integration,
- academic and linguistic institutions for language validation and community engagement,
- technology partners for platform development and scaling,
- and national AI ecosystem partners such as BharatGen and other language-AI initiatives.

The program may also explore structured mechanisms for long-term sustainability, deployment management, and ecosystem partnerships through an appropriate institutional vehicle.

9. Expected Outcomes

At the end of three years, AdiVaani Phase-II is expected to deliver:

- multilingual AI support for a substantially expanded set of tribal languages,
- translation pipelines connecting tribal, state, and national languages,

- speech-to-speech communication systems,
- OCR and digitization pipelines for educational and policy materials,
- translated and digitized school textbooks and public documents,
- searchable archives of folktales, oral narratives, and cultural heritage,
- a Tribal Language LLM enabling generative AI functionalities,
- and sectoral AI tools for education, governance, healthcare, and media.

10. Concluding Position

AdiVaani Phase-II should be positioned not as a narrow translation project, but as:

A national AI ecosystem for tribal languages, knowledge access, educational inclusion, cultural preservation, and last-mile digital empowerment.

Such a program will create the foundation for a multilingual and inclusive AI future in which India's linguistic diversity is not treated as a barrier, but as a strategic national strength.